

LESSON 125 (1.0 Ground and 1.0 AATD)**Lesson Objective**

The purpose of this lesson is to introduce maneuvering solely by reference to flight instruments and introduce basic radio navigation techniques and recoveries from unusual attitudes to facilitate aeronautical decision making and situational awareness.

Briefing

- Outline the objectives and completion standards of the lesson.
- Establish the student and instructor responsibilities.
- Review homework assigned in the previous lesson.
 - Quiz the student on the readings as needed:
 - Chapter 06 of the Airplane Flying Handbook - Takeoffs and Departure Climbs:
 - Short field takeoffs.
 - Soft field takeoffs.
 - Chapter 08 of the Airplane Flying Handbook - Airport Traffic Patterns:
 - Short field landings.
 - Soft field landings.
 - Warrior's Maneuvers' Manual:
 - Short field takeoffs.
 - Soft field takeoffs.
 - Short field landings.
 - Soft field landings.
- Student questions.

References

- Airplane Flying Handbook. ([FAA-H-8083-3C](#))
 - Chapter 05 - Maintaining Aircraft Control: Upset Prevention and Recovery Training.
- Instrument Flying Handbook. ([FAA-H-8083-15B](#))
 - Chapter 06, Section 1 - Airplane Attitude Instrument Flying Using Analog Instrumentation.
 - Chapter 06, Section 2 - Airplane Attitude Instrument Flying Using an Electronic Flight Display.
 - Chapter 07, Section 1 - Airplane Basic Flight Maneuvers Using Analog Instrumentation.
 - Chapter 07, Section 2 - Airplane Basic Flight Maneuvers Using an Electronic Flight Display.
 - Chapter 09 - Navigation Systems.
- Pilot's Handbook of Aeronautical Knowledge. ([FAA-H-8083-25B](#))
 - Chapter 16 - Navigation.
- Aeronautical Information Manual. ([AIM](#))
 - Chapter 01 - Air Navigation.

Lesson Content (Ground)

Basic Instrument Flying Techniques

- Introduce why pilots need to know these techniques even if flying VFR.
 - Inadvertent VFR flight into instrument meteorological conditions (IMC).
 - This occurs when the airplane cannot be controlled with reference to the natural horizon.
 - This can happen due to improper preflight planning, un-forecasted weather, and operating in marginal VFR conditions.
 - Necessary steps to take when encountering IMC:
 - Do not panic.
 - Recognize the need for immediate action.
 - Begin a scan inside with your primary instrument being the attitude indicator.
 - Maintain positive control of the airplane.
 - Level the wings and stop the climb/descent.
 - Trust what the flight instruments are showing.
 - Do not trust your senses!
 - Contact ATC and get assistance.
 - They can identify nearby obstructions, guide you back to VFR conditions, talk to you through an approach, etc.
 - Exit IMC.
 - The best way is to conduct a 180-degree turn. Turn back to where you know it was VFR conditions!
 - A descent can be an option as well if you are sure that it is VFR below.
 - If altitude allows.
 - Be careful of terrain and obstacles.
- Fundamental skills of basic attitude instrument (BAI).
 - Instrument cross-check (Scan).
 - Purpose is to prevent disorientation, unusual attitudes, and recognition of a failed instrument.
 - There are different types of instrument scanning methods. The radial scan is preferred.
 - Use one of the panel posters in the briefing rooms to demonstrate the radial scan method.
 - Attitude indicator → airspeed indicator → attitude indicator → altimeter → attitude indicator → HSI → attitude indicator → RPM → attitude indicator → turn coordinator → attitude indicator.
 - Introduce: fixation, omission, and emphasis.
 - Instrument interpretation.
 - Determine the aircraft's attitude and performance based on the instrument scan.
 - Aircraft control.

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- Apply the required control responses to fly the airplane.
- Control and performance method:
 - While flying in IMC, the control and performance method is to be used.
 - Pitch + Power = performance.
 - Setting a certain pitch with a power setting results in the desired performance.
 - For example: around 2.5 degrees of pitch + around 2300 RPM = straight and level flight at 90 knots.
 - For example: around 8-10 degrees of pitch + full RPM = climb at 79 knots.
- VOR and GPS navigation procedures:
 - Refer to the VOR and GPS sections in the lesson content section below.

Unusual Attitude Recoveries

- Loss of control in-flight (LOC-I): a significant deviation of an aircraft from an intended flight path usually resulting in an airplane upset.
 - This can occur in any phase of flight.
 - The FAA defines an upset as “an event that unintentionally exceeds the parameters normally experienced in flight or training” (FAA, 2021).
 - Pitch attitude greater than 25°, nose up.
 - Pitch attitude greater than 10°, nose down.
 - Bank angle greater than 45°.
 - Within the above parameters but flying at airspeeds inappropriate for the conditions.
- Recovery from an unusual attitude:
 - Nose down:
 - Power idle.
 - Prevents excessive airspeed and loss of altitude.
 - Wings level.
 - Wings must be level before pitching up → cover what happens when load factor increases.
 - Pitch up.
 - Return to level flight attitude.
 - Nose up:
 - Full power.
 - Prevents getting too slow and stalling.
 - Wings level.
 - In a nose up attitude, the bank and pitch can be adjusted simultaneously.
 - Pitch down.
 - In a nose up attitude, the bank and pitch can be adjusted simultaneously.

Lesson Content (AATD)

Introduction to Attitude Indicator

- Define what each line on the attitude indicator represents.
 - Degrees of pitch.
 - In increments of 2.5°.
 - Blue represents positive pitch.
 - Brown represents negative pitch.
 - Degrees of bank.
 - 10, 20, 30, 45, 60
 - 30 and 60 are extended lines.
 - Airspeed.
 - Altitude.
 - Standard rate.
 - Inclinator.
 - Red chevrons.
 - When you exceed 30 degrees of pitch.

Attitude Flying

- Refer to the fundamental skills of basic attitude instrument and the control and performance method sections above.
 - Instrument errors:
 - Cover how to recognize an instrument error/failure and how to solve it
 - ADAHRS failure:
 - Red X's.
 - Reset the circuit breaker (has to be done within 30 seconds of the AHRS failure).
 - If PFD function does not return then use the standby instruments, contact ATC, and land as soon as practical.
 - DAU failure (Avidyne):
 - Red X's.
 - Reset the circuit breaker.
 - If MFD function does not return then contact ATC, and land as soon as practical.
 - Pitot-static blockages:

Situation / Instrument	Airspeed	Altimeter	VSI
Blocked pitot	Zero	Works	Works
Blocked pitot and drain hole	High in climb Low in descent	Works	Works

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Blocked static	Low in climb High in descent	Frozen	Frozen
Alternate static	Reads high	Reads high	Momentary climb

- Straight and level:
 - Have the student maintain straight and level flight at 90 kts.
 - Pitch + Power = performance.
 - Around 2.5 degrees of pitch + around 2300 RPM = straight and level flight at 90 knots.
 - If a higher or lower airspeed is desired at a straight and level setting, the pitch and power settings will have to be adjusted.
 - Lower pitch attitude + higher RPM = higher airspeed.
 - Higher pitch attitude + lower RPM = lower airspeed.
 - Control instruments: attitude indicator, HSI, tachometer, and turn coordinator.
 - Performance instruments: airspeed indicator, altimeter, and VSI.
 - Emphasize the need to scan!
- Descending:
 - Have the student perform a glide descent.
 - Pitch + Power = performance.
 - Nose right under the horizon line (attitude indicator) + power idle = glide descent at 73 knots.
 - Practice until proficient.
 - If a higher or lower airspeed is desired, the pitch and power settings will have to be adjusted.
 - For example, a descent at 90 knots is maintained at around 1600-1800 RPM and about -2.5 degrees of pitch.
 - Leveling off should commence at about 10% of the VSI.
 - Pitch, power, trim still applies!
 - Control instruments: attitude indicator, HSI, tachometer, and turn coordinator.
 - Performance instruments: airspeed indicator, altimeter, and VSI.
 - Emphasize the need to scan!
- Climbing:
 - Have the student perform a climb at V_Y .
 - Pitch + Power = performance.
 - About a 10 degrees pitch attitude + full power = climb at 79 knots.
 - Practice until proficient.
 - Have the student perform a climb at V_{climb} .
 - About a 5-7.5 degrees of pitch attitude + full power = climb at 87 knots.
 - Leveling off should commence at about 10% of the VSI.

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- Pitch, power, trim still applies!
 - Control instruments: attitude indicator, HSI, tachometer, and turn coordinator.
 - Performance instruments: airspeed indicator, altimeter, and VSI.
 - Emphasize the need to scan!
- Standard rate turn:
 - A standard rate turn is defined as a 3 degrees per second turn.
 - A 180 degrees turn takes 1 minute and 360 degrees turn takes 2 minutes.
 - To calculate how many degrees of bank a standard rate of turn is, use the following formula:
 - Divide airspeed by 10 and add $\frac{1}{2}$ of the result.
 - Warrior standard rate:
 - $90/10 = 9$.
 - $9/2 = 4.5$.
 - $4.5 + 9 = 13.5$ degrees = standard rate.
 - Have the student perform turns at a standard rate. The bank should be in between the 10 (1st mark) and 20 (2nd mark) degrees mark on the attitude indicator.
 - Remind the student that as the bank increases, pitch must increase to compensate for the reduction in vertical lift.
 - Verify that the student keeps the airplane coordinated.
 - Emphasize the need to scan!
- Medium bank turns:
 - A medium bank turn must be performed at 30 degrees of bank.
 - Have the student perform turns at 30 degrees of bank (3rd bank mark on the attitude indicator).
 - Remind the student that as the bank increases, pitch must increase to compensate for the reduction in vertical lift.
 - Verify that the student keeps the airplane coordinated.
 - Emphasize the need to scan!

VOR Navigation Procedures

- Demonstrate how to use the VOR.
 - Set the VOR frequency in the G430 (show where to find the frequency).
 - Tune and ID the VOR by pressing the ID button and NAV1 or NAV2 (depending on which NAV is being used).
 - Verify with the morse code on a sectional chart.
 - For the G5:
 - Once tune and ID is complete, press the CDI button on the G430 to switch it from GPS to VLOC.
 - The G5 HSI should then show the green VOR needle.
 - To adjust the course, press the left knob, scroll to the right to CRS, and press the left knob. This allows the pilot to twist to the desired course.

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- For the Avidyne:
 - Once tune and ID is complete, press the up-most left bezel key on the PFD. This will cycle between GPS1, VLOC1, GPS2, and VLOC2. Select the desired VLOC.
 - To adjust the course, use the left knob on the PFD to adjust the needle to the desired course. To center it, simply press the left knob.
- Cover the difference between an outbound and inbound indication.
 - Outbound: arrows point away from each other.
 - Inbound: arrows point the same way.
 - When given an outbound instruction like “N609FT intercept and track the 360-radial outbound”, the course should be set to 360.
 - When given an inbound instruction like “N609FT intercept and track the 360-radial inbound”, the tail should be set to 360 OR take the reciprocal of 360 and set that number to the course.
 - For the purpose of this lesson, the student should be able to:
 - Figure out the current radial.
 - Intercept and track inbound/outbound instructions using an intercept angle of 30-45 degrees in the correct direction.

GPS Navigation Procedures

- Demonstrate how to use the GPS.
 - Show the different sections of the G430.
 - NAV, waypoint, AUX, and nearest.
 - Demonstrate the various ways of proceeding direct to an airport.
 - Direct button → input the airport identifier → enter → enter.
 - Nearest page → nearest airports → direct → enter → enter.
 - For the G5:
 - Press the CDI button on the G430 to switch it from VLOC to GPS.
 - The G5 HSI should then show the magenta GPS needle.
 - For the Avidyne:
 - Press the up-most left bezel key on the PFD. This will cycle between GPS1, VLOC1, GPS2, and VLOC2. Select the desired GPS.
 - For the purpose of this lesson, the student should be able to:
 - Proceed direct to an airport.
 - Intercept and track a GPS course.

Unusual Attitudes

- Put the student in an unusual attitude and verify that they can recover properly.
- Recovery from an unusual attitude:
 - Nose down:
 - Power idle.
 - Prevents excessive airspeed and loss of altitude.
 - Wings level.

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- Wings must be level before pitching up → cover what happens when load factor increases.
- Pitch up.
 - Return to level flight attitude.
- Nose up:
 - Full power.
 - Prevents getting too slow and stalling.
 - Wings level.
 - In a nose up attitude, the bank and pitch can be adjusted simultaneously.
 - Pitch down.
 - In a nose up attitude, the bank and pitch can be adjusted simultaneously.

Post-Brief

- Student self-critique.
- Instructor critique.
- Questions from the student.
- Lesson grading.
- Logbook completion.
- Schedule the next lesson.
- Preview the next lesson → 126.
- Assign homework:
 - Review BAI and other items from this lesson.
 - Review chapter 06 of the Airplane Flying Handbook - Takeoffs and Departure Climbs:
 - Short field takeoffs.
 - Soft field takeoffs.
 - Review chapter 08 of the Airplane Flying Handbook - Airport Traffic Patterns:
 - Short field landings.
 - Soft field landings.

Completion Standards

This lesson is complete when the student is able to, with instructor guidance:

1. Use radio aids for orientation.
2. Track and bracket radials.
3. Identify and practice proper corrective actions to unusual attitudes.
4. Maintain altitude ± 200 ft, heading $\pm 30^\circ$, and airspeed ± 10 kts.